Stem-and-Leaf Plots

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Introduction

The stem-and-leaf plot, introduced by John Tukey, is a compact display that shows the distribution of a small sample while preserving every data value. Each observation is split into a "stem" (leading digits) and a "leaf" (trailing digit); stems are listed vertically and leaves are written next to each stem in order. For samples up to about 100 observations, the resulting ASCII plot is a poor man's histogram that can be typed into a notebook.

Prerequisites

The reader should know what a histogram is.

Theory

Given observations, pick a stem width (10s, 100s, etc.) and list each observation by its stem and leaf. For example, values 12, 15, 18, 22, 25, 29, 34 with tens-digit stems:

```
1 | 2 5 8
2 | 2 5 9
3 | 4
```

The back-to-back variant puts two groups on either side of the stem, permitting visual comparison. Modern software rarely uses stem-and-leaf for publication figures, but it remains a quick inspection tool in R's console.

Assumptions

Purely descriptive, requires nothing except numeric data.

R Implementation

```r
set.seed(2026)
x <- round(rnorm(40, mean = 100, sd = 15))
stem(x)

# With scale change
stem(x, scale = 0.5)
stem(x, scale = 2)

# Back-to-back
library(aplpack)
x <- rnorm(30, 100, 15)
y <- rnorm(30, 110, 12)
stem.leaf.backback(x, y)
```

Output & Results

```
  The decimal point is 1 digit(s) to the right of the |

   7 | 3
   8 | 0447
   9 | 01135789
  10 | 0111345899
  11 | 0245578
  12 | 34
```

Each leaf on the 10-row represents one observation: 101, 101, 101, 103, 104, etc. The display is itself a histogram with the same-height bars rotated 90 degrees.

Interpretation

Use stem-and-leaf plots for quick eyeballing in R's console; they reveal outliers, skew, and multimodality without generating a figure file.

Practical Tips

- For samples over 100, the plot becomes cluttered; use a histogram.
- `scale` in `stem()` adjusts the stem width; experiment to find the right granularity.
- Back-to-back plots are a useful low-overhead alternative to side-by-side boxplots.
- Stem-and-leaf is rarely used in journal articles today; it is a Tukey EDA tradition, not a reporting convention.
- In teaching contexts, it remains a useful way to introduce distribution shape before histograms.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Descriptive statistics and Table 1